



PRS Pricing Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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David K Kelly

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1 Test Results — PRS Pricing Model (MKM-PR-001)

Tests run on **03 April 2026 at 20:15:57**.

Table 1: Test Results Summary — PRS Pricing Model (MKM-PR-001)

Total Tests	57
Passed	42
Failed	15
Skipped	0
Duration	1.75s

Table 2: Detailed Test Results — PRS Pricing Model

Test Description	Acceptance Criteria	Result	Time
1Y from Feb 2026 = last Friday May 2027.	<code>mat == last_friday_of_month(2027, 5)</code>	Pass	0.000s
3Y from Feb 2026 = last Friday May 2029.	<code>mat == last_friday_of_month(2029, 5);</code> <code>mat.month == 5 (+1 more)</code>	Pass	0.000s
Jan 2027 uses Aug 2026 roll → Nov 2029 maturity.	<code>mat == last_friday_of_month(2029, 11)</code>	Pass	0.000s
3Y from Sep 2026 = last Friday Nov 2029 (Aug roll).	<code>mat == last_friday_of_month(2029, 11);</code> <code>mat.month == 11</code>	Pass	0.000s
5Y from Feb 2026 = last Friday May 2031.	<code>mat == last_friday_of_month(2031, 5)</code>	Pass	0.000s
5Y from Sep 2026 = last Friday Nov 2031.	<code>mat == last_friday_of_month(2031, 11)</code>	Pass	0.000s
All maturities should fall on a Friday.	<code>mat.weekday() == 4</code>	Pass	0.000s
3Y maturity jumps from May to Nov at the Aug roll.	<code>may_mat.month == 5; nov_mat.month == 11 (+1 more)</code>	Pass	0.000s
Aug onwards uses Aug roll.	<code>roll == last_friday_of_month(2026, 8);</code> <code>roll.month == 8</code>	Pass	0.000s
December uses the same year's Aug roll.	<code>roll.month == 8; roll.year == 2026</code>	Pass	0.000s
Feb-Jul uses Feb roll.	<code>roll == last_friday_of_month(2026, 2);</code> <code>roll.month == 2</code>	Pass	0.000s
January uses the previous year's Aug roll.	<code>roll == last_friday_of_month(2026, 8);</code> <code>roll.month == 8 (+1 more)</code>	Pass	0.000s
July is the last month of the Feb roll period.	<code>roll.month == 2; roll.year == 2026</code>	Pass	0.000s
March is still in the Feb roll period.	<code>roll.month == 2; roll.year == 2026</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
compute_maturity_date with no ref_date uses today.	isinstance(mat, date); mat.weekday() == 4	Pass	0.000s
current_roll_date(None) uses today and returns a valid date.	isinstance(roll, date); roll.weekday() == 4	Pass	0.000s
maturity_table with no ref_date uses today.	len(table) == 5; all(isinstance(e['maturity_date'], date) for e in table)	Pass	0.000s
Empty propertyts dir	stats["total_properties"] == 0; stats["properties_with_gev"] == 0	FAIL	0.010s
Generator should still work without gauge hazard curves (empty basis).	stats["properties_with_gev"] == 2; stats["properties_with_floor"] == 1 (+1 more)	FAIL	0.066s
No propertyts dir	—	Pass	0.000s
Aug 2026: 31st is Monday, last Friday = 28th.	last_friday_of_month(2026, 8) == date(2026, 8, 28)	Pass	0.000s
Feb 2026: 28th is Saturday, last Friday = 27th.	last_friday_of_month(2026, 2) == date(2026, 2, 27)	Pass	0.000s
May 2029: 31st is Thursday, last Friday = 25th.	last_friday_of_month(2029, 5) == date(2029, 5, 25)	Pass	0.000s
Nov 2029: 30th is Friday.	last_friday_of_month(2029, 11) == date(2029, 11, 30)	Pass	0.000s
Result should always be a Friday (weekday=4).	d.weekday() == 4	Pass	0.000s
Actual years should be strictly increasing.	years[i] > years[i - 1]	Pass	0.000s
All maturities should be in May for the Feb roll.	e['maturity_date'].month == 5	Pass	0.000s
All maturities should be in November for the Aug roll.	e['maturity_date'].month == 11	Pass	0.000s
Maturity string should be formatted as 'DD Mon YYYY'.	len(e['maturity_str']) > 8; 'May' in e['maturity_str'] or 'Nov' in e['maturity_str']	Pass	0.000s
Default table has 5 entries (1Y through 5Y).	len(table) == 5	Pass	0.000s
Tenors should be 1, 2, 3, 4, 5.	tenors == [1, 2, 3, 4, 5]	Pass	0.000s
Positive spread	spread > 0; 50 < spread < 500	FAIL	0.000s
Spread increases with hazard	spread_high > spread_low	FAIL	0.000s
Spread values at different tenors	s > 0	FAIL	0.000s
Zero hazard rate	spread == 0.0	FAIL	0.000s
Basis calculation	len(nearest) > 0; ng0["gauge_id"] == "GAUGE-001" (+5 more)	FAIL	0.070s

Test Description	Acceptance Criteria	Result	Time
Depth thresholds	"any_flood" in thresholds; "moderate" in thresholds (+4 more)	FAIL	0.070s
Generate produces output	output_path.exists(); "metadata" in data (+2 more)	Pass	0.067s
Gev params structure	"shape" in gev; "loc" in gev (+2 more)	FAIL	0.069s
High transmission rate	prop3["summary"]["flood_transmission_rate"] == 1.0	Pass	0.070s
Output structure	"PROP-001" in curves; "PROP-002" in curves (+14 more)	FAIL	0.067s
Properties with gev	stats["properties_with_gev"] == 2; stats["properties_with_floor"] == 1 (+1 more)	FAIL	0.065s
Survival decreases with tenor	survival[i] <= survival[i - 1]	FAIL	0.070s
Term structure	ts["tenors"] == TENORS; threshold_name in ts (+3 more)	FAIL	0.070s
Recovery rates by trigger	len(spreads) > 0	FAIL	0.071s
Recovery reduces spread	spread_with_recovery < spread_no_recovery	FAIL	0.000s
After attach_spread_decomposition, propertyhc.json has decomposition.	"spread_decomposition" in pc; "gauge_spread_bps" in sd (+5 more)	Pass	0.288s
At least one effect per path must be non-zero when gauge != property.	df["distance_effect_bps"] != 0.0 or df["elevation_effect..."; ef["elevation_effect_bps"] != 0.0 or ef["distance_effect...]	Pass	0.209s
Both decomposition paths should arrive at the property spread.	abs(path1 - prop) < 0.1; abs(path2 - prop) < 0.1	Pass	0.206s
With shd/she files present, decomposition values must be non-zero.	sd["property_spread_bps"] != 0.0; sd["shd_spread_bps"] != 0.0 (+1 more)	Pass	0.209s
Without shd/she files, decomposition still works (zeros).	n > 0; sd["shd_spread_bps"] == 0.0 (+1 more)	Pass	0.070s
_get_5yr_spread must return index 4 when available.	PropertyHazardCurveGenerator._get_5yr_spread == 50.0	Pass	0.000s
_get_5yr_spread must return 0.0 for empty/missing data.	PropertyHazardCurveGenerator._get_5yr_spread == 0.0; PropertyHazardCurveGenerator._get_5yr_spread(... (+1 more)	Pass	0.000s
_get_5yr_spread must work with specified threshold.	PropertyHazardCurveGenerator._get_5yr_spread('severe'...; PropertyHazardCurveGenerator._get_5yr_spread(pc, 'any_flo...	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All tenors should round-trip through maturity and back.	<code>tenor_from_maturity(mat, ref) == t</code>	Pass	0.000s
A date not in the table returns the closest tenor.	<code>result in [1, 2, 3, 4, 5]</code>	Pass	0.000s
Exact maturity date returns correct tenor.	<code>tenor_from_maturity(mat, date(2026, 2, 24)) == 3</code>	Pass	0.000s

Failed Test Details

test_empty_propertyts_dir File: tests/port/propertyhc/basis.py

```
1 tests/port/propertyhc/basis.py:233: in test_empty_propertyts_dir
2     assert stats["properties_with_gev"] == 0
3 E   KeyError: 'properties_with_gev'
```

test_no_gauge_hazard_curves File: tests/port/propertyhc/basis.py

```
1 tests/port/propertyhc/basis.py:240: in test_no_gauge_hazard_curves
2     assert stats["properties_with_gev"] == 2
3 E   KeyError: 'properties_with_gev'
```

test_properties_with_gev File: tests/port/propertyhc/generator.py

```
1 tests/port/propertyhc/generator.py:54: in test_properties_with_gev
2     assert stats["properties_with_gev"] == 2
3 E   KeyError: 'properties_with_gev'
```

test_output_structure File: tests/port/propertyhc/generator.py

```
1 tests/port/propertyhc/generator.py:73: in test_output_structure
2     assert prop1["has_gev"] is True
3 E   AssertionError
```

test_gev_params_structure File: tests/port/propertyhc/generator.py

```
1 tests/port/propertyhc/generator.py:95: in test_gev_params_structure
2     assert "shape" in gev
3 E   TypeError: argument of type 'NoneType' is not iterable
```

test_depth_thresholds File: tests/port/propertyhc/generator.py

```
1 tests/port/propertyhc/generator.py:108: in test_depth_thresholds
2     assert "any_flood" in thresholds
3 E   AssertionError
```

test_term_structure File: tests/port/propertyhc/generator.py

```
1 tests/port/propertyhc/generator.py:127: in test_term_structure
2     assert threshold_name in ts
3 E   AssertionError
```

test_survival_decreases_with_tenor File: tests/port/propertyhc/generator.py

```

1 tests/port/propertyhc/generator.py:140: in test_survival_decreases_with_tenor
2     survival = ts["any_flood"]["survival"]
3 E   KeyError: 'any_flood'

```

test_basis_calculation File: tests/port/propertyhc/generator.py

```

1 tests/port/propertyhc/generator.py:161: in test_basis_calculation
2     assert ng0["event_basis"] == 15
3 E   AssertionError

```

test_zero_hazard_rate File: tests/port/propertyhc/pricing.py

```

1 tests/port/propertyhc/pricing.py:38: in test_zero_hazard_rate
2     spread = gen._compute_prs_spread(0.0, 5, None)
3 E   AttributeError: 'PropertyHazardCurveGenerator' object has no attribute '_compute_prs_spread'

```

test_positive_spread File: tests/port/propertyhc/pricing.py

```

1 tests/port/propertyhc/pricing.py:43: in test_positive_spread
2     spread = gen._compute_prs_spread(0.02, 5, None)
3 E   AttributeError: 'PropertyHazardCurveGenerator' object has no attribute '_compute_prs_spread'

```

test_spread_increases_with_hazard File: tests/port/propertyhc/pricing.py

```

1 tests/port/propertyhc/pricing.py:49: in test_spread_increases_with_hazard
2     spread_low = gen._compute_prs_spread(0.01, 5, None)
3 E   AttributeError: 'PropertyHazardCurveGenerator' object has no attribute '_compute_prs_spread'

```

test_spread_values_at_different_tenors File: tests/port/propertyhc/pricing.py

```

1 tests/port/propertyhc/pricing.py:55: in test_spread_values_at_different_tenors
2     spreads = [gen._compute_prs_spread(0.02, t, None) for t in [1, 5, 10, 30]]
3 E   AttributeError: 'PropertyHazardCurveGenerator' object has no attribute '_compute_prs_spread'

```

test_recovery_reduces_spread File: tests/port/propertyhc/pricing.py

```

1 tests/port/propertyhc/pricing.py:65: in test_recovery_reduces_spread
2     spread_no_recovery = gen._compute_prs_spread(0.03, 5, None, recovery=0.0)
3 E   AttributeError: 'PropertyHazardCurveGenerator' object has no attribute '_compute_prs_spread'

```

test_recovery_rates_by_trigger File: tests/port/propertyhc/pricing.py

```

1 tests/port/propertyhc/pricing.py:78: in test_recovery_rates_by_trigger
2     spreads = ts[name]["prs_spread_bps"]
3 E   KeyError: 'any_flood'

```

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (30 tests)	D. Kelly
07-Mar-2026	1.1	57 test(s) added; 30 test(s) removed (total: 57)	D. Kelly
07-Mar-2026	1.2	30 test(s) added (total: 87)	D. Kelly
18-Mar-2026	1.3	4 test(s) added (total: 91)	D. Kelly
27-Mar-2026	1.4	30 test(s) removed (total: 61)	D. Kelly
30-Mar-2026	1.5	3 test(s) added; 12 test(s) removed (total: 52)	D. Kelly
01-Apr-2026	1.6	4 test(s) added (total: 56)	D. Kelly
03-Apr-2026	1.7	3 test(s) added; 2 test(s) removed (total: 57)	D. Kelly